

INPUT: 7 FcμR-Fcμ complexes
PDB: 7YTE,7YTC,7YTD,7YSG,8BPE,8BPF,8BPG
mIgM -> 2:1 pIgM -> 1:1/4:1 (same side) & 8:1 (both sides) sIgM -> 4:1 (opposite SC)

STRUCTURAL ALIGNMENT
· Pairwise Cα RMSD (MDAnalysis)
· Superposition (ChimeraX)

CONTACT ANALYSIS
· Residue contacts (Biopython)
· Contact proportion heatmaps
· Interaction types (PLIP/PDBePISA)

EVOLUTION & ALANINE SCANNING
· ConSurf conservation (MAFFT)
· ΔΔG predictions (FoldX+PremPS)
· In silico mutation (PyMOL) & H-bond (ChimeraX)

MULTI-CRITERIA FILTERING
(structural, evolutionary, energetic, functional occurrence)

PRIMARY ENERGETIC HOTSPOT
Arg45
· ΔΔG >1 (both platforms)
· ConSurf 9, full salt bridges

STRUCTURAL ANCHOR
Phe67
· Hydrophobic packing (5/5)
· Moderate conservation (grade 6)
· ΔΔG method-dependent >1

STRUCTURAL SUPPORTING HOTSPOTS
Thr60, Ser63, Thr110, Asp111
· Highly conserved (grades 8-9)
· Form H-bonds/salt bridges (5/5)
· ΔΔG inconsistent

COOPERATIVE HOTSPOT
Lys69
· Near-zero ΔΔG alone
· Synergistic with Phe67 (double mutant abolishes binding)

NON-ESSENTIAL RESIDUES
Thr65, Asn66, Asn109, Arg112
· Low conservation (grades 3-4)
· Infrequent contacts, weak ΔΔG

UNIFIED HIERARCHICAL MODEL
· Rigid lock-and-key interface (Cα RMSD 1.13-2.07 Å)
· Stoichiometric plasticity without binding-mode change
· IgM-exclusive Cμ4 network explains isotype specificity

THERAPEUTIC BLUEPRINT
Structure-guided targeting of primary energetic and cooperative hotspots for CLL, SLE and B-cell malignancies